PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze data.	**Python, SQL**	SQL is used for data extraction and analytical querying from relational databases, while Python (Pandas/NumPy) is utilized for data cleaning, statistical validation, and missing-value imputation across 5,000 patient records.
2.	**Database / Data Warehouse**	Warehouse Used to store and manage structured business data for analysis and reporting.	**CSV File / Excel Data Source**	Provides a lightweight, portable, and structured flat-file datastore containing all 5,000 patient clinical records (BMI, Blood Pressure, Glucose, Cholesterol) and lifestyle metrics (Smoking, Alcohol, Exercise, Fatigue) for fast extraction.
3.	**Data Ingestion**	Used to load data from source files or external systems into the analytical environment.	**Tableau Data Connections**	Establishes a direct live/extract connection from flat-file datasets (CSV/Excel) into Tableau Desktop, enforcing automated schema mapping and field type assignments for clinical indicators and demographic variables.
4.	**Data Transformation and Business Logic**	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	**Tableau Prep / Tableau Desktop Calculated Fields**	Executes business logic to bin patient ages into 5 age groups (Teenager to Old), categorize BMI levels into 4 risk tiers (Obese: 39.98%, Overweight: 32.91%, Normal, Underweight), and compute population health KPI measures (Diabetes Rate: 34.92%, Heart Disease Rate: 29.98%, Stroke Rate: 24.86%).
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, charts, reports, KPIs,	**Tableau Desktop**	Renders 8 visual worksheets and interactive story points, including

		and visual analysis.		Blood Pressure vs. Glucose scatter plots, Age Disease heatmaps, Exercise-level pie charts (Levels 0–2), and Smoking status comparative bar charts.
6.	**Front-End / User Interface**	Interface through which users interact with dashboards, filters, parameters, and visual insights.	**Tableau Public / Tableau Web View**	Delivers a responsive, web-accessible interactive dashboard portal allowing healthcare stakeholders (Dr. Michael Carter, Emily Rodriguez, Dr. Priya Nair) to dynamically filter, explore, and stratify patient risk.